

Aubin Birba

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Cybersecurity engineer and PhD candidate specialized in applied cryptography, with expertise in PKI, key management, and secure protocol design for high-assurance environments. Experienced in bridging research and practical implementation, delivering privacy-preserving solutions for cloud, distributed systems and machine learning, and integrating encryption services into critical applications.

Education

Jan. 2022 – **Université du Québec À Montréal (UQÀM), Canada**
Present

PhD in Computer Science - Applied Cybersecurity and Cryptography in Collaborative Machine Learning. Supervised by Prof. Marc-Olivier Killijian and Prof. Sébastien Gambs.

- Engineered Diffie-Hellman protocol variants for secure key exchange in constrained trust environments (OpenSSL).
- Designed privacy-preserving protocols securing sensitive data in distributed networks (e.g., encrypted recommendation engines).
- Re-engineered cryptographic protocols with homomorphic encryption to optimize for cloud-based workloads (OpenFHE).
- Built decision tree training protocol on encrypted datasets with secure unlearning support (TFHE-rs).
- Developed encrypted cloud storage and search platform ensuring confidentiality and access control.

Oct. 2018 – **Dual Degree – Computer Engineering & MSc in Cyber-Defense and Information Security, UPHF / ENSA El Jadida, Valenciennes, France**
Oct. 2021

Focused on offensive/defensive security, vulnerability management, cryptography, and compliance in critical sectors (finance, industry).

Oct. 2016 – **Undergraduate Diploma (DEUG) – Mathematics and Computer Science,**
Sept. 2018 *Faculty of Sciences, El Jadida, Morocco*

Professional Experience

2024 – **Lecturer – Penetration Testing & Cyber Defense, Champlain College, Canada**
Present

- Designed and delivered hands-on courses covering network security, application security, PKI, key management, and cryptographic protocols.
- Created structured, standards-aligned documentation and technical procedures inspired by NIST CSF.
- Led real-world lab sessions on secure application development, OWASP Top 10 mitigation, and penetration testing.
- Applied Agile teaching methodology to drive collaboration and practical problem-solving.

- 2023 - **Teaching Assistant**, *UQÀM*, Montreal, Canada
- Present ○ Guided students in secure systems design, Linux administration, and secure coding (Java, web frameworks).
○ Mentored teams on documenting cryptographic processes and drafting technical requirements.
- Mar. – Sept. 2021 **Junior Security Engineer Intern – Application Security**, *Expleo Group*, Montigny-le-Bretonneux, France
○ Developed SDLC-integrated security methodology for critical environments (aerospace, automotive, rail), achieving compliance with standards such as IEC 63452.
○ Built compliance support tools (Odoos, Python, Security Knowledge Framework, C/C++) that streamlined developer adherence to security policies.
- Jul. – Sept. 2020 **Junior Developer Intern**, *Z Smart Services*, Marrakech, Morocco
○ Created a scientific data visualization platform with automated data extraction and normalization scripts.
○ Implemented secure storage and data handling protocols, reducing exposure to potential breaches.

Skills

Cryptography & Data Security

- RSA, AES, digital signatures, PKI, homomorphic encryption, secure multi-party computation.
- Key management, secrets management, applied cryptographic protocol design.
- OWASP Top 10 vulnerability assessment.
- Network security : Snort, Suricata, Splunk, Sentinel

Development & Systems

- Languages: Python, Java, C++, Rust, JavaScript, TypeScript.
- Frameworks: Angular, Spring Boot, Django, Laravel.
- Databases: MySQL, PostgreSQL, MongoDB, Firebase.
- Systems: Linux administration, Git, Docker.

Secure Machine Learning

- ML on encrypted data, secure unlearning, adversarial training, outlier detection
- Tools: TensorFlow, scikit-learn, TFHE-rs, OpenFHE, SPDZ, OpenSSL

Methodologies

- Agile/Scrum, technical documentation, secure SDLC integration, research and scientific writing.

Languages

French (native), English (professional proficiency).

Certifications

- Jul. 2025 **CompTIA Security+**, *CompTIA*
- Asset/threat/vulnerability identification, risk management.
 - Intrusion detection/prevention, applied cryptography.
- Jul. 2025 **Google Cybersecurity Professional Certificate**, *Google*
- Security fundamentals, network security, Linux, MySQL.
 - Threat detection/response, automation.